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The transformation of vaterite to calcite: effect of the conditions of the solutions in contact with the mineral phase

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Abstract

The transformation of the thermodynamically unstable vaterite into the stable calcite was investigated during the precipitation of calcium carbonate, under conditions of constant supersaturation at 25, 35 and 45°C and pHs 8.5, 9.0 and 10.0. The calcite content in the solid precipitates was determined quantitatively by the X-ray diffraction technique (XRD) using the appropriate mixed standards of known composition. Scanning electron microscopy (SEM) and particle number and size measurements were also used for the characterisation of the solid precipitates. It was found that the transformation of vaterite into calcite does not depend on pH and temperature in the range 25–45°C. It depends only on the supersaturation, and this, only at relatively high supersaturation ratios (1.5–1.9). Specifically, the rate of transformation decreases as supersaturation increases. It was suggested on the basis of the kinetics results, that the transformation takes place through dissolution of vaterite, preferably of the small crystals followed by the crystallisation of calcite. At relatively high supersaturation ratios (1.5–1.9) the transformation is controlled by the dissolution of vaterite, whereas at lower supersaturation ratios (1.2–1.5) the rate of dissolution of vaterite is similar with that of crystallisation of calcite. Finally, a model equation which predicts a linear dependence of the rate of transformation on supersaturation ratio has been derived at relatively high supersaturation ratios. © 1998 Elsevier Science B.V. All rights reserved.

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1. Introduction

The precipitation and dissolution processes of calcium carbonate are very important in a vast number of scientific areas and relevant research is relentlessly continuing, since, several aspects contributing to our understanding of these processes remain still unclear. One of the features which is of key importance to the interpretation of phenomena related to precipitation and dissolution processes is the polymorphism of carbonate [1]. The polymorphs which are formed at ambient temperature and in the absence of additives during the

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spontaneous or seeded crystal growth of calcium carbonate include vaterite and calcite [2–4]. Depending on the temperature of the supersaturated solutions and on the level of concentration of foreign ions present aragonite may also be stabilized [5]. The formation of transient polymorphic phases which may be stabilized may strongly affect the properties of the precipitates because of the marked differences not only in the physicochemical properties but also because of the difference in the morphological characteristics of the various polymorphs.

In the present work we have investigated the process of transformation of the thermodynamically unstable vaterite into the least soluble polymorph, calcite, by carefully monitoring the transformation at conditions of constant supersaturation. The present work supplements other studies concerning vaterite transformation into calcite [1–4], since by keeping the supersaturation constant we were able to investigate quantitatively with high precision the parameters which are most important for this process. Thus, the effect of pH over the range 8.5–10.0 and of the temperature at 25, 35 and 45°C was investigated. From the kinetics measurements at these conditions a model was proposed describing the kinetics of vaterite transformation into calcite.

2. Experimental procedure

The experiments were done in double walled, water jacketed, thermostated pyrex vessel, volume totalling $V = 0.2\ dm^3$ at 25, 35 and 45 and pHs 8.5, 9.0 and 10.0. Stock calcium nitrate and sodium nitrate solutions were prepared from the respective solids (Merk, Puriss) using triply distilled water. Sodium bicarbonate stock solutions were prepared fresh for each experiment from the solid reagent (Riedel de Haen AG). Equal volumes ($0.1\ dm^3$) of calcium nitrate and sodium bicarbonate prepared from the respective stock solutions were simultaneously added (mixing time ca. 15 s) into the reaction vessel. Immediately following the mixing the pH of the supersaturated solutions was adjusted through the addition of standard potassium hydroxide. The pH measurement, calibration of the glass/saturated calomel electrode pair used, and stock solution standardisation have been described in detail elsewhere [6,7]. Following pH adjustment, the solutions were stable as verified by the constancy of the solution pH. Following the lapse of easily measurable and reproducible (better than 10%, mean of four experiments) induction times, precipitation accompanied by proton release in the bulk solution started. A decrease in the solution pH as small as 0.003 pH units triggered the addition of titrant solutions of calcium nitrate and sodium carbonate having the stoichiometry (1 : 1) of the precipitating $CaCO_3$. The additions were done from the two mechanically coupled burettes of a modified automatic titrator controlled by a personal computer. The composition of the titrant solutions, calculated so as to maintain the solution supersaturation throughout the course of the precipitation experiments, has been described in detail elsewhere [8]. The supersaturation at the onset of the precipitation (initial supersaturation, S_{in}) was relatively high and was kept constant for approximately 1 h. Next, the titrants added were replaced with standard potassium hydroxide solution which was added for a short time during which desupersaturation proceeded to the desired extent which was determined through preliminary experiments. Following the attainment of the desired relative supersaturation, this supersaturation was also kept constant by the addition of calcium nitrate and sodium carbonate titrants of the appropriate concentrations. Following this procedure a lower supersaturation (final supersaturation, S_f) was maintained for a long time and the pH was kept constant through the entire course of the precipitation process (initial, decreasing and final supersaturation). The steps involved are shown in Fig. 1.

During the course of the precipitation process, samples were withdrawn so as to keep approximately a constant volume and were filtered through membrane filters (Millipore 0.2 μm). The filtrates were analysed for calcium by atomic absorption spectroscopy in order to confirm the constancy of the solution composition. In all experiments the analysis showed that during the crystallisation process, calcium concentration remained constant to within $\pm$ 3%. The solids precipitated after drying at 120°C for 1 h were characterised by powder

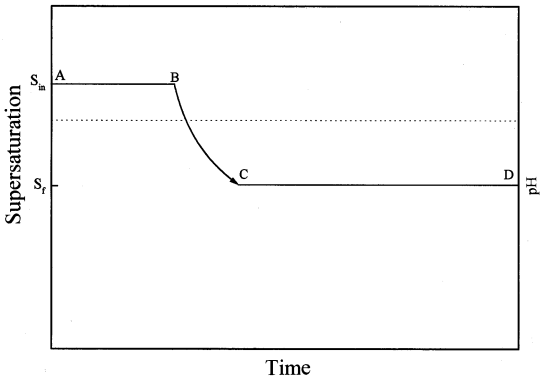


Fig. 1. Sequence of experimental conditions for the investigation of the kinetics of vaterite transformation at constant supersaturation AB: titrants composition to maintain S_{in}; BC: desupersaturation; CD: titrants composition to maintain S_f ($S_{in} > S_f$): (——) supersaturation as a function of time; (- - -) pH as a function of time.

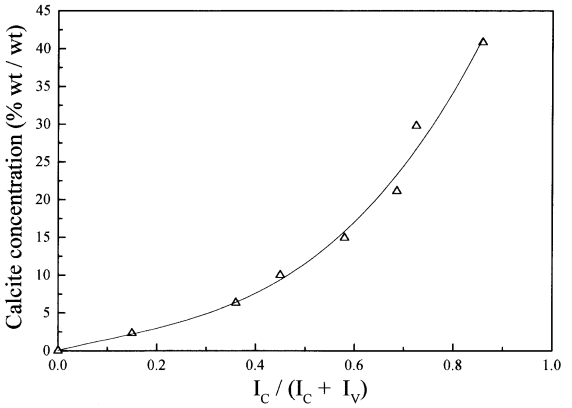


Fig. 2. Quantitative analysis of calcite–vaterite binary mixtures by powder X-ray diffraction. Plot of the calcite content in the precipitate as a function of the ratio of the relative intensities $I_c/I_c + I_v$ for the 104 calcite peak, I_c, and the 110 vaterite peak I_v. Calibration curve.

X-ray diffraction (XRD, Philips 1830/40) and scanning electron microscopy (SEM, JEOL ISM 5200). The particle number, particle size and size distribution of the precipitate were measured by a counter based on a rotating laser beam diffraction (Spectrex ILI 1000).

The calcite content in the solid precipitates was determined by the powder XRD technique [9–12]. The height of the characteristic diffraction peaks of calcite, I_c, and vaterite, I_v, corresponding to the 1 0 4 and 1 1 0 reflections respectively were used for the calculation of the ratio $I_c/(I_c + I_v)$. The reproducibility was better than 7%, mean of three measurements. The conversion of this ratio to weight percent of calcite was done by means of a calibration curve, which was constructed by mixing pure calcite and pure vaterite in different proportions, measuring the $I_c/(I_c + I_v)$ ratio and plotting these values against weight percent calcite. A typical calibration curve is shown in Fig. 2.

3. Results

The driving force for the formation of a calcium carbonate polymorph, x, is defined as the change in Gibbs free energy for going from the supersaturated solution to equilibrium:

$$\Delta G_x = -R_g T \ln S_x, \quad (1)$$

where R_g is the gas constant, T the absolute temperature and S_x the supersaturation ratio

$$S_x = \{\mathrm{IAP}/\mathrm{IAP}_{eq,x}\}^{1/2}, \quad (2)$$

where IAP and IAP_{eq} are the ionic activity products in the supersaturated solution and in equilibrium with the polymorph x, respectively. The computation of the supersaturation ratio was done by using the computer code HYDRAQL [13] which is a free energy minimization program taking into account all equilibria in the solution, mass balance and electroneutrality conditions. With respect to the complicating gas/liquid equilibrium of carbon dioxide it was neglected in all cases using large solution volumes, high pH and minimal air volume above the aqueous phase [14]. The physicochemical characterisation (XRD and SEM) of the solid precipitates showed that the thermodynamically unstable vaterite formed during the initial supersaturation as well as in the beginning of the final supersaturation, in accordance with Refs. [3,15–17]. The computations were therefore done with respect to vaterite. The computed supersaturation ratios and the corresponding temperatures and pH values used in this study are summarized in Table 1.

Table 1
Initial, S_{in}, and final, S_f, supersaturation ratios used at different temperatures and pH values and the respective rates of transformation of vaterite into calcite, r_{tr}. (calcite content is expressed as weight % in the total precipitate mass)

t (°C)	pH	S_{in}	S_f	$r_{tr}/m_{calcite}$ (h^{-1})
25	8.5	2.55	1.42	2.91
25	8.5	2.55	1.50	3.12
25	8.5	2.55	1.75	1.10
25	9.0	2.81	1.20	2.44
25	9.0	2.81	1.28	2.33
25	9.0	2.81	1.43	2.78
25	9.0	2.81	1.79	0.63
25	10.0	4.10	1.45	2.81
25	10.0	4.10	1.60	1.68
25	10.0	4.10	1.84	0.33
35	9.0	2.56	1.41	2.41
45	9.0	2.55	1.44	2.89

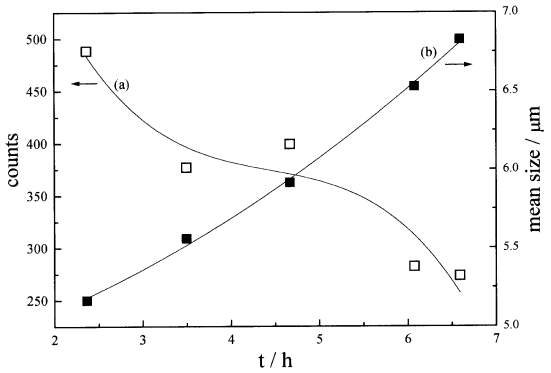


Fig. 3. Evolution of the number (curve a) and their mean size (curve b) of precipitated $CaCO_3$ crystallites during the final supersaturation. $S_f = 1.79$, pH = 9.0, $t = 25.0$°C.

Fig. 3 shows the variation of the number (curve a) of $CaCO_3$ crystallites and their mean size (curve b) with time, by considering as time zero the start of the final supersaturation. Curves a and b are typical profiles of the particles development for all experiments in this work. It may be seen that the number of the particles decreased with time whereas their size increased.

Typical scanning electron micrographs shown in Fig. 4 show the morphology of the solid

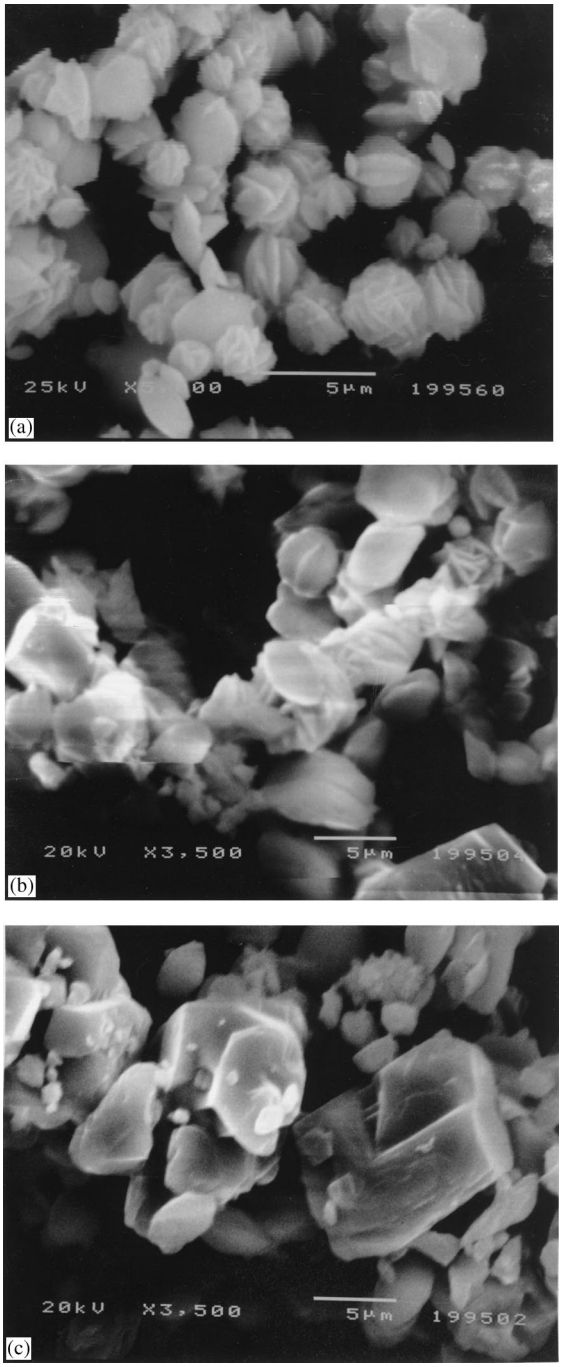


Fig. 4. Typical scanning electron micrographs of solid precipitates withdrawn from the reactor in the beginning (a), at $t = 75$ min (b) and at $t = 450$ min (c) during the final supersaturation. $S_f = 1.20$, pH = 9.0 and $t = 25$°C.

precipitates withdrawn from the reaction vessel in the beginning (a) and at two different times (b, c) during the time in which the final supersaturation was kept constant. The micrographs show that pure vaterite exists exclusively in the beginning of the final supersaturation (micrograph a) and a gradual transformation of the thermodynamically unstable spherical vaterite into the stable cubic calcite takes place during the final supersaturation (micrographs b and c) over the entire range of conditions investigated.

4. Discussion

As mentioned in the experimental section, quantitative analysis of the vaterite–calcite content in the solid precipitates was done using the appropriate calibration curve. The analytical results are shown in Fig. 5 by considering as zeroth time the start of the final supersaturation. A linear dependence of calcite content as a function of time was found at all conditions investigated. The slope of the straight lines obtained from linear regression of the calcite content as a function of time was taken as the rate of transformation of vaterite into calcite, r_{tr}. The values calculated from the respective plots are compiled in the last column of Table 1. From the plots

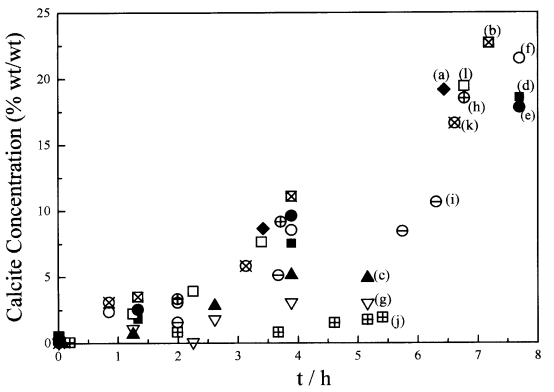


Fig. 5. Calcite content in the precipitate as a function of time (i) pH = 8.5, $t = 25°C$ and $S_f = 1.42$ (a), 1.50 (b), 1.75 (c); (ii) pH = 9.0, $t = 25°C$ and $S_f = 1.20$ (d), 1.28 (e), 1.43 (f), 1.79 (g); (iii) pH = 10.0, $t = 25°C$ and $S_f = 1.45$ (h), 1.60 (i), 1.84 (j); (iv) pH = 9.0, $t = 35°C$, $S_f = 1.41$ (k) and (v) pH = 9.0, $t = 45°C$, $S_f = 1.44$ (l).

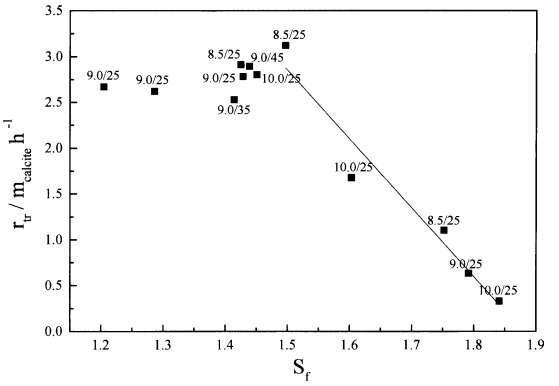


Fig. 6. Variation of the transformation rate of vaterite into calcite versus the final supersaturation ratio. The experimental conditions (pH and temperature) are quoted at each point.

of the transformation rates as a function of the corresponding final supersaturation ratio (Fig. 6, at each point, the respective pH and temperature in °C are quoted) it may be observed that for relatively low supersaturation ratios, 1.2–1.5, the transformation rate remained constant whereas for higher supersaturation ratios, 1.5–1.9, the transformation rate decreased significantly as the supersaturation increased. Examination of the rates corresponding to points obtained at the same temperature and approximately equal supersaturations but at different pH values i.e. of the points labelled as 8.5/25, 9.0/25 and 10.0/25 corresponding to supersaturation ratios 1.42, 1.43 and 1.45, respectively, leads to the conclusion that pH does not affect the transformation of vaterite into calcite. Moreover, by observing the rates corresponding to points obtained at the same pH and approximately equal supersaturations but at different temperatures, i.e. the points labelled as 9.0/25, 9.0/35 and 9.0/45 with supersaturation ratios 1.43, 1.41 and 1.44, respectively, it may be inferred that in the range 25–45°C, temperature does not affect the transformation of vaterite into calcite, as well. So the only factor which seems to affect the aforementioned transformation is the solution supersaturation in which the solid phase is formed.

These results may be formulated into a model which describes the transformation of vaterite into calcite. Fig. 7 shows the solubility (IAP_{eq}) of

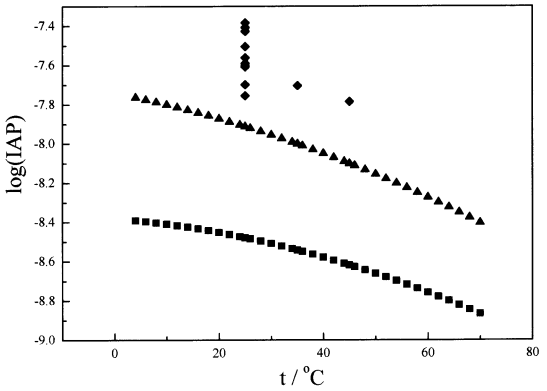


Fig. 7. Ionic activity product of the working solutions (diamonds), and of the saturated solutions in equilibrium with vaterite (triangles) and calcite (squares) as a function of temperature.

vaterite and calcite as well as the IAP of the working solutions used in this study. It is evident that the working solutions are supersaturated with respect to both vaterite and calcite and that vaterite is the metastable (more soluble) phase whereas calcite is the stable (less soluble) phase. The fact that vaterite forms initially demonstrates that the formation of vaterite is favoured kinetically under the present experimental conditions. With reference to the gradual transformation of vaterite into calcite it may be suggested that the more soluble vaterite crystals dissolve while the less soluble calcite crystals nucleate and grow [18]. It should be noticed here that the solid-state transformation must be eliminated because vaterite following drying is not transformed into another phase for a long time period (of several months). The only possible mechanism therefore of transformation of vaterite into calcite, is the solution-mediated mechanism as described before. The driving force for this transformation, ΔG_{tr}, is given by

$$\Delta G_{tr} = \Delta G_c - \Delta G_v, \tag{3}$$

where ΔG_c and ΔG_v are the driving force for the formation of calcite and vaterite, given by Eq. (1). By combining Eqs. (1)–(3) we may express ΔG_{tr} as a function of the solubilities of vaterite and calcite.

$$\Delta G_{tr} = -\frac{R_g T}{2} \ln \frac{IAP_{eq,v}}{IAP_{eq,c}}. \tag{4}$$

A similar equation has been derived by Turnbull [19,20] for the calculation of the change in Gibbs free energy of the transformation of vaterite into calcite.

Eq. (4) does not predict any dependence of the driving force and then of the rate of transformation, r_{tr}, on the supersaturation ratio, contrary to our experimental results (Fig. 6). In order to explain this contradiction as well as the dependence of r_{tr} on the supersaturation ratio (illustrated in Fig. 6) we assumed that at relatively high supersaturations the dissolution of vaterite is a very slow process compared with the crystallisation of calcite. Thus, it is possible that the dissolution of vaterite controls the transformation of vaterite into calcite, in agreement with Yamaguchi and Murakawa [21] and the rate of transformation equals that of dissolution. The driving force of formation and hence of the stability of vaterite (ΔG_v) is given by Eq. (5) [22]:

$$\Delta G_v = -RT_g \ln S_v. \tag{5}$$

Growth and dissolution processes are saturation dependent processes. The higher the supersaturation, the higher will be the tendency for the formation of vaterite rather than its dissolution and subsequent transformation into calcite. It is therefore anticipated that the rate of vaterite transformation will increase with decreasing supersaturation ratio. This may be seen from Eq. (5). If we compare the driving force for the stabilization of vaterite at two supersaturation values S_{v1} and S_{v2} where $S_{v1} > S_{v2}$

$$(\Delta G_v)_1 = -R_g T \ln S_{v_1},$$

$$(\Delta G_v)_2 = -R_g T \ln S_{v_2},$$

$$\Delta(\Delta G_v) = (\Delta G_v)_2 - (\Delta G_v)_1 = -R_g T \ln \frac{S_{v_2}}{S_{v_1}}. \tag{6}$$

According to Eq. (6) the relative stability of vaterite at the lower supersaturation will be less favourable since the ratio $S_{v_2}/S_{v_1} < 1$.

The plot of the rate of vaterite transformation into calcite expressed as a function of the supersaturation ratio with respect to vaterite in the solutions is linear as shown in Fig. 6, over the range of supersaturation ratios between 1.5 and 1.9. A model equation describing the kinetics of transformation of vaterite into calcite in the range of relatively high supersaturation ratios obtained from linear least squares fit is

$$r_{tr} = -7.51S_f + 14.12. \quad (7)$$

Considering now the range of relatively low supersaturation ratios (1.2–1.5) the observed independence of r_{tr} from S_f may be explained according to our model as follows: In accordance with Eqs. (1) and (5), as S_f decreases the driving force for dissolution of vaterite becomes more favourable [Eq. (5)] and thus the rate of dissolution of vaterite increases, whereas the driving force for the formation of calcite becomes less favourable [Eq. (1)] and thus the rate of crystallisation of calcite decreases. It is therefore anticipated that a threshold of supersaturation ratio exists below which the rate of dissolution of vaterite is comparable with the rate of crystallisation of calcite. Past this supersaturation threshold the transformation rate is not controlled by the dissolution of vaterite. In this case the driving force of transformation, given by Eq. (4) and the rate of transformation of vaterite into calcite is independent from supersaturation ratio. According to our experimental results (Fig. 6) the aforementioned threshold of S_f must be of value 1.5. As may be seen the range of supersaturation ratios in which the rate of vaterite transformation into calcite can be varied depending on the composition of the supersaturated solutions is very narrow.

Finally the decreasing number and the increasing mean size of particles with time, illustrated in Fig. 3, demonstrate the preferable dissolution to the small particles of vaterite.

Our experimental results have been interpreted by adopting the dissolution of vaterite as the rate determining step for the transformation of vaterite into calcite at relatively high supersaturation ratios (1.5–1.9). The disagreement of our results with those obtained by Ogino et al. [23,24] who proposed that the rate-determining step is the crystallisation of calcite, may be attributed to the fact that these researchers studied the transformation of vaterite to calcite in a suspension where the supersaturation ratio with respect to vaterite was equal to unity, that is much lower than that in our case. According to our model, decreasing the relative supersaturation with respect to vaterite increases its rate of dissolution whereas the rate of crystallisation of calcite decreases. It is therefore possible, in solutions saturated with respect to vaterite the crystallisation rate of calcite becomes predominant.

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